Carson Manuel Montini

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Education

California Baptist University, Riverside, CA — Expected Apr 2027
Bachelor of Science in Mechanical Engineering | ABET-accredited Program | Physics Minor
Cumulative GPA: 3.33

Technical Skills

Design: SolidWorks | Fusion 360 | AutoCAD | GD&T | Design for Manufacturing | Iterative Design
Analysis: Finite Element Analysis (FEA) | Computational Fluid Dynamics (CFD) | MATLAB/Simulink | Topology
Manufacturing: CNC Machining | CAM | Mills | Lathes | Welding | Additive Manufacturing | Carbon-fiber Fabrication
Programming and Tools: C++ | C# | Python | Arduino | Microsoft Excel | Microsoft Office | Spanish Biliteracy

Engineering Experience

Battery & Electromechanical Systems | Formula SAE Electric Team — Sep 2023 – July 2026
California Baptist University, Riverside, CA

- Designed the accumulator enclosure, segment packaging, connector interfaces, and precision components in SolidWorks for safe, serviceable vehicle integration.
- Reduced battery weight 12% and accumulator volume 20% through structural and packaging optimization while preserving safety and manufacturability requirements.
- Designed and evaluated nickel busbar fuse geometries using fuse-width calculations, transient thermal analysis, and infrared testing to characterize current-interruption behavior.
- Coordinated mechanical and electrical interfaces across 10 vehicle subsystems and prepared component designs for CNC and additive manufacturing using design-for-manufacturing practices.

Fuselage and Landing Gear Subsystems | SAE Aero Design Team — Jul 2026 – Present
California Baptist University, Riverside, CA

- Develop fuselage and landing gear concepts for a competition aircraft, balancing payload capacity, structural support, weight, and manufacturability.
- Coordinate fuselage packaging and component placement with the team to accommodate a secured payload and maintain flightworthy empty and loaded center of gravity.
- Design landing gear geometry and mounting surfaces for takeoff and landing loads, ground clearance, and positive ground steering.

Selected Engineering Projects

Automated Court Enhancer | Multidisciplinary Team Project — Nov 2025 – Apr 2026
California Baptist University, Riverside, CA

- Designed and assembled a 12-inch autonomous hardwood-court cleaning robot integrating dual DC drive motors, a rotating cleaning pad, fluid pump and misters, and onboard water storage.
- Sized mechanical and electrical components against 1.0-1.2 m/s speed and 2-hour runtime design targets; developed 12 V power distribution and 5 V control architecture.
- Integrated Arduino Mega and H-bridge controls to coordinate drive, cleaning, and fluid-delivery functions within a compact chassis.

Reverse-Engineered Industrial Clamp | Additive Manufacturing Team Project — Sep 2023 – Nov 2023
California Baptist University, Riverside, CA

- Reverse-engineered an industrial clamp in SolidWorks and developed a finite element model to evaluate stress distribution, deformation, and critical load paths.
- Design an enclosed fuselage cargo bay for secured 2L bottle payloads, with access for competition time limits
- Conducted mesh-refinement and convergence studies to verify solution stability, then compared simulation trends with three-point bending and tensile-test results from Markforged carbon-fiber and fiberglass reinforced specimens.

Leadership Experience

Resident Advisor — Aug 2024 – Apr 2026
California Baptist University, Riverside, CA

- Mentored residents, planned community programs, communicated university policies, and responded to conduct and safety concerns in a residential setting.